Feature Importance for Applied Research

From model inspection to statistical inference

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Why feature importance?

- Feature *selection*: which candidate predictors carry most signal?
- *Sanity* check: does a model rely on plausible signal?
- *Hypothesis* generation
- *Inference* on the data-generating process
- **Caveat**: FI ≠ causal effect, but associations are still informative (Ewald et al. 2024)

> **!** “Importance” is not a single quantity → *method choice* defines the answer

Which features matter?

Term	Estimate	SE	p
`hour`	7.8	0.4	< 0.001
`humidity`	−197.9	16.4	< 0.001
`temperature`	351	22.5	< 0.001
`windspeed`	18.5	22.6	0.412

`lm(count ~ .)` on bike-sharing data (more later)

- Linear model: compare $|\hat{\beta}_j|$, t-test p-values
- Random forest, XGBoost, neural net: no β to read
- Feature importance (FI) generalizes into that gap



But: Important *for what*, and *how* is it measured?

Importance ≠ effect

Effect

- "What happens to $\hat{y}$ *when* X_j *changes*?"
- Linear model: β_j
- *Signed*, in units of X_j
- For arbitrary prediction methods:
  - Partial dependence (*PDP*)
  - Accumulated local effects (*ALE*)

Importance

- "*How much* does the model / learner rely on X_j?"
- Linear model: $|\beta_j|$ (eyeballing), t-test
- Unsigned (generally), one number per feature
- Extends to nonlinear effects + interactions (depending on learner)
- Scale depends on the loss → read as a *ranking*

Taxonomy & Estimands

A rough taxonomy of interpretibility methods

Today: model-agnostic, loss-based, global

Model-specific
(G)LM coefficients, Gini impurity, gradient-based

vs.

Model-agnostic
supports any learner

Prediction-based
How much do predictions change without a feature?

vs.

Loss-based
how much “worse” does the model get without a feature?

Local
One observation’s prediction (e.g., SHAP)

vs.

Global
the model as a whole, feature-specific

What are we estimating?

Which question?

- **Marginal** ("crude"): $X_j \not\perp\!\!\!\perp Y$
  Is X_j related to Y at all?
- **Conditional** ("adjusted"): $X_j \not\perp\!\!\!\perp Y \mid X_{-j}$
  Does X_j add information *given the others*?

Which target?

- **Model**: this one fitted model
- **Learner**: the fitting procedure, across datasets
- **Data-generating process** (Williamson et al. 2023)

⚠ Most methods estimate *model* importance, research questions usually target the *data*
Linear model: t-test for β_j is the closest analogue, not the same thing

Feature Importance Methods

Why not just refit?

Leave-one-covariate-out (LOCO)

Calculation

1. Fit model $\hat{f}$ with all features
2. Refit $\hat{f}_{-j}$ without X_j
3. Compare test loss

$$\text{LOCO}_j = L\left(\hat{f}_{-j}\right) - L\left(\hat{f}\right)$$

- Effectively: one step of backward selection
- Groups: leave-one-*group*-out (LOGO)
- Leave-one-covariate-*in* (LOCI): univariate association
- *Cost*: one refit per feature → expensive for
  - Complex algorithms / tuning
  - Huge datasets

📝 **Test loss, e.g. mean squared error**

$L\left(\hat{f}\right) = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{f}(x_i)\right)^2$ → importance is on the scale of the *loss*

Don’t refit, perturb

Permutation feature importance (PFI)

Calculation

1. Fit model $\hat{f}$ once
2. Shuffle X_j in the test data: $\tilde{x}_j$
3. Compare test loss before and after

$$\text{PFI}_j = L\left(\hat{f}, \tilde{x}_j\right) - L\left(\hat{f}\right)$$

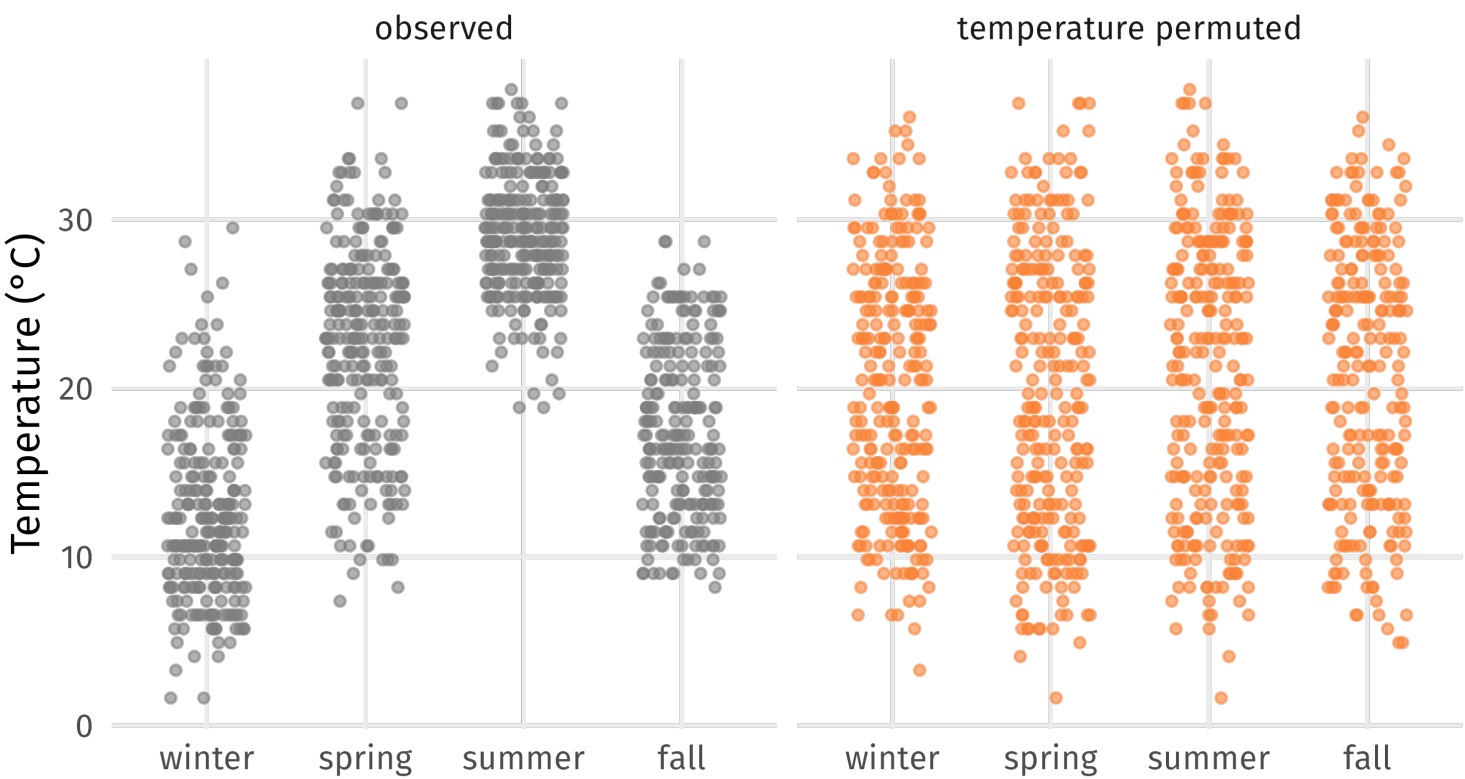
4. Repeat k times for stability

- No refits, works with any learner
- Breaks the link to Y *and* to the other features
- Correlated features → implausible combinations
  - Ex. A 20-year-old with 30 years of smoking history

Shuffling leaves the data

Model is evaluated where it has never seen data

(Hooker et al. 2021)



Perturb, but keep the dependence

Conditional feature importance (CFI)

- Sample replacement values $\tilde{x}_j$ from $X_j \mid X_{-j}$ instead of marginal shuffling
- Removes only the part of X_j *not* explained by the others
- Stays on the data manifold
- Otherwise: Same procedure as PFI:

$$\text{CFI}_j = L\left(\hat{f}; \tilde{x}_j\right) - L\left(\hat{f}\right), \quad \tilde{x}_j \sim X_j \mid X_{-j}$$

! Needs samples from $X_j \mid X_{-j}$, which are not easy to get

Between PFI and CFI

Relative feature importance (RFI)

(König et al. 2021)

Sample X_j from $X_j \mid X_G$ for a chosen *conditioning set* G

$G = \varnothing$

- PFI
- "Does the model use X_j?"

$G = \{\text{age}, \text{sex}\}$

- RFI
- "Beyond age and sex?"

$G = -j$

- CFI
- "Beyond all other features?"

Where do conditional samples come from?

- *Gaussian*: fast, continuous features only
- *Knockoff* framework (Candès et al. 2018)
- Tabular data is usually *mixed*: continuous, categorical, missing values

💡 **Adversarial random forests (ARF)**

- Generative model for tabular data (Watson et al. 2023)
- Mixed types, missing values
- Sampler for CFI and RFI in `{xplainfi}` (Blesch et al. 2025)

Is it more than noise?

Inference for CFI

- CFI is a mean of per-observation loss differences, $\tilde{x}_i$: x_i with x_{ij} resampled

$$\Delta_i = \left(y_i - \hat{f}(\tilde{x}_i)\right)^2 - \left(y_i - \hat{f}(x_i)\right)^2, \quad \text{CFI}_j = \overline{\Delta} = \frac{1}{n}\sum_{i=1}^{n} \Delta_i$$

- One-sided paired test on $\overline{\Delta}$ (e.g., t-test, Wilcoxon)
  - *CPI*: conditional predictive impact (Watson and Wright 2021)
  - *cARFi*: same test with ARF sampling (Blesch et al. 2025)

💡 Holdout set, model fixed
→ *model* importance

📝 Many features:
Control FWER or FDR
(Holm, Benjamini–Hochberg, ...)

And now for some game theory

Shapley Additive Global importancE (SAGE)

- Value of a feature set S: loss reduction when only features in S are known

$$v(S) = L(f_\varnothing) - L(f_S)$$

- **Shapley value**: average gain from adding j (Shapley 1953)

$$\varphi_j = \sum_{S \subseteq D \setminus \{j\}} \frac{|S|!(p - |S| - 1)!}{p!} [v(S \cup \{j\}) - v(S)]$$

- Unknown features: MC integration over marginal or conditional samples, like PFI vs. CFI

📝 **SAGE vs. SHAP**

SAGE: *global*, explains the loss
(Covert et al. 2020)

SHAP: *local*, explains one prediction
(Lundberg and Lee 2017)

Application Example

The bike sharing dataset

Capital Bikeshare, Washington, D.C., 2011–2012

- *Outcome*: bike rentals per hour
- 3000 hours sampled from ~17k
- Random forest (`ranger`) and `lm`
- Holdout: 80% train, 20% test

9 features

- **Calendar**: hour, weekday, working day, season, year
- **Weather**: temperature, humidity, wind speed, weather type

```
library(xplainfi)
library(mlr3learners)

task <- tsk("bike_sharing")
learner <- lrn("regr.ranger", num.trees = 1000)
measure <- msr("regr.mse")
resampling <- rsmp("holdout", ratio = 0.8)
```

PFI


```
pfi <- PFI$new(
  task, learner, measure,
  resampling = resampling,
  n_repeats = 10
)
pfi$compute()
pfi$importance()
```

Feature	PFI
hour	43665
temperature	9034
working_day	7140
year	6537
humidity	2061
season	1022
weather	510
weekday	420
windspeed	122
holiday	30

CFI


```
cfi <- CFI$new(
  task, learner, measure,
  resampling = resampling,
  n_repeats = 10,
  sampler = ConditionalARFSampler$new(task)
)
cfi$compute()
cfi$importance()
```

Feature	CFI
hour	38821
year	5877
temperature	5490
working_day	2457
humidity	1315
season	824
weather	348
weekday	301
windspeed	87
holiday	28

LOCO


```
loco <- LOCO$new(
  task, learner, measure,
  resampling = resampling
)
loco$compute()
loco$importance()
```

Feature	LOCO
hour	19596
year	3272
temperature	1506
working_day	1281
weather	272
season	210
humidity	198
weekday	55
holiday	13
windspeed	-45

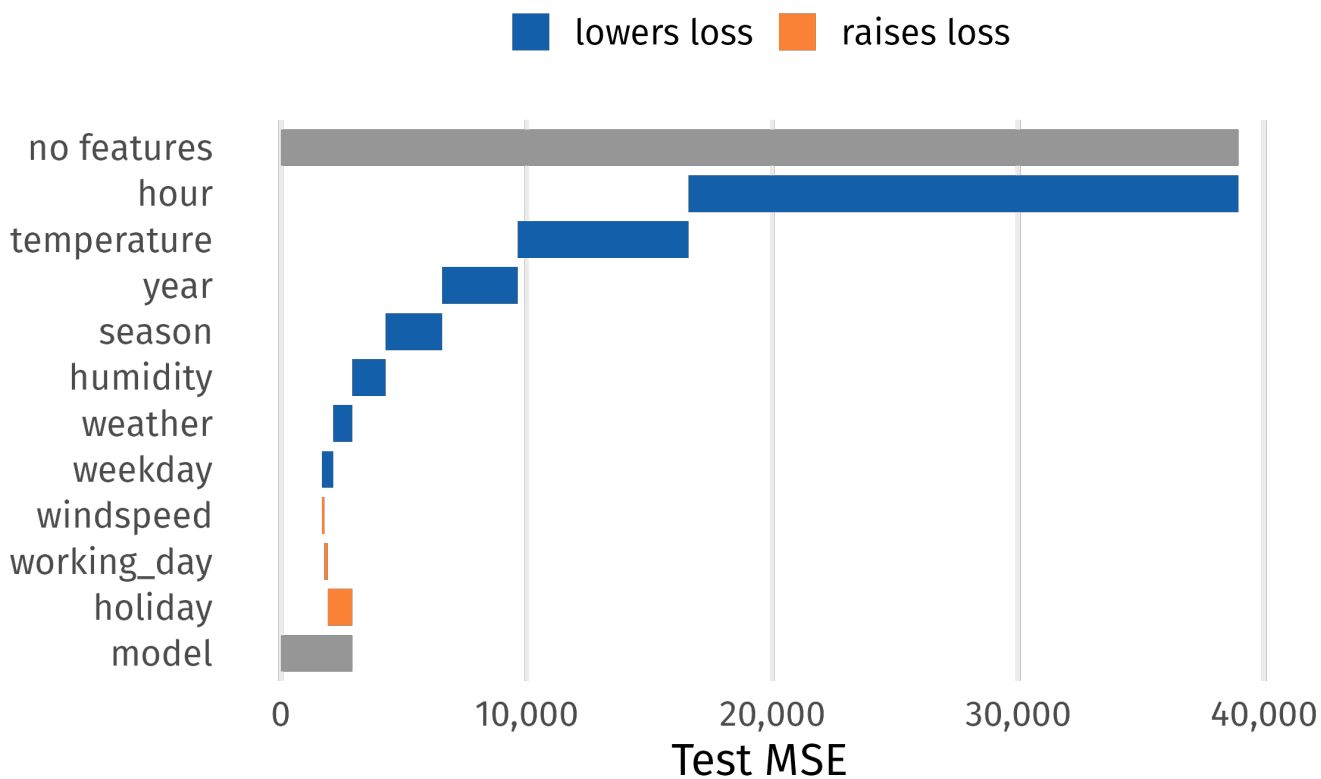
SAGE


```
sage <- MarginalSAGE$new(
  task, learner, measure,
  resampling = resampling,
  estimator = "kernel",
  n_coalitions = 16384
)
sage$compute()
sage$importance()
```

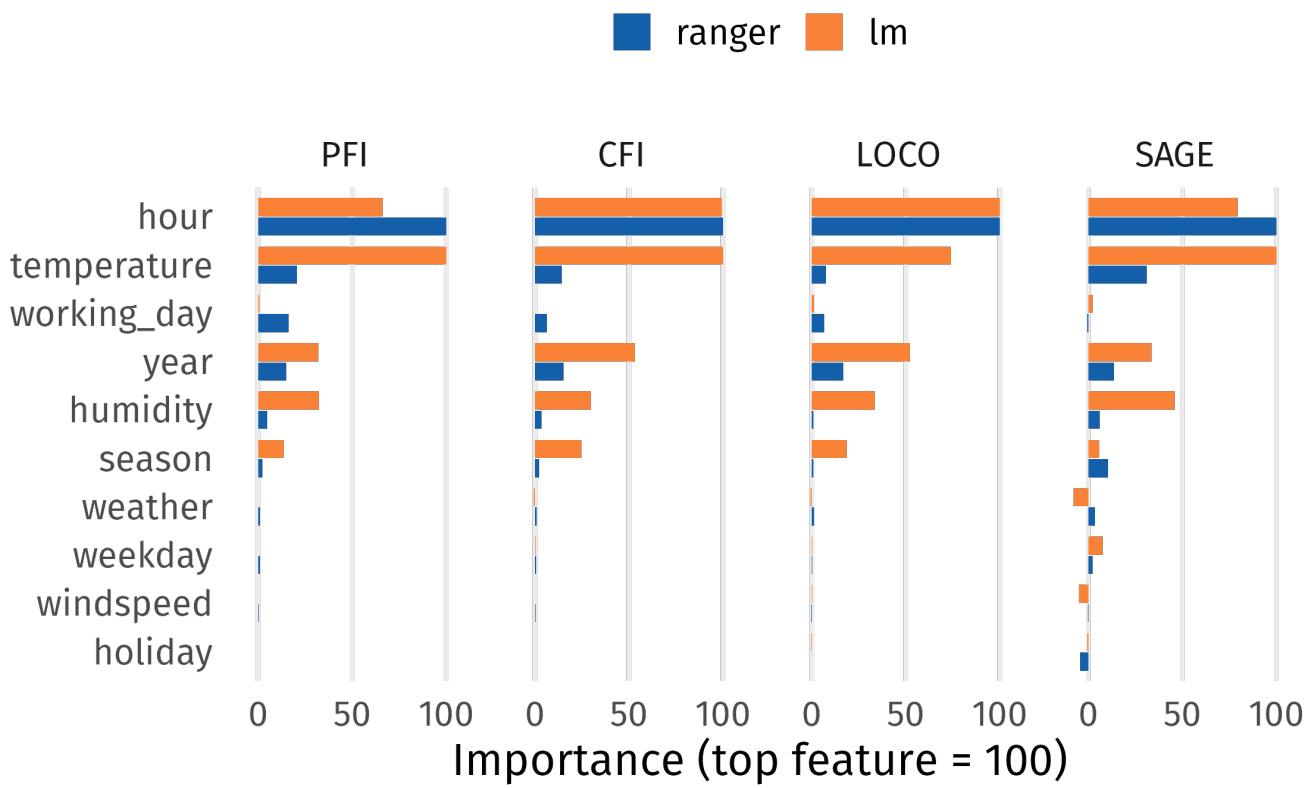
Feature	SAGE
hour	22382
temperature	6956
year	3059
season	2311
humidity	1351
weather	777
weekday	469
windspeed	-111
working_day	-146
holiday	-981

SAGE adds up

Loss without features → loss of the model, one step per feature



Comparing methods and learners



Inference

CFI with CPI, 600 held-out hours


```
cfi$importance(ci_method = "cpi", p_adjust = "BH")
```

Feature	CFI	SE	Lower 95%	p (BH)
hour	38821	2150	34598	< 0.001
year	5877	544	4808	< 0.001
temperature	5490	675	4164	< 0.001
working_day	2457	340	1789	< 0.001
humidity	1315	244	837	< 0.001
season	824	132	565	< 0.001
weather	348	84	183	< 0.001
weekday	301	55	193	< 0.001
windspeed	87	56	-22	0.119
holiday	28	26	-24	0.284

- `hour`, `year`, `temperature` clearly important
- `windspeed`, `holiday`: no evidence

Summary

Question	Method	Watch out for
Does the model use X_j?	PFI	extrapolation
Information beyond the others?	CFI: perturb	conditional sampler
	LOCO: refit	refits, learner tuning
Share of the explained loss?	SAGE	cost, estimator noise

- CFI and LOCO: near-identical estimands, different estimation (Ewald et al. 2024)

! Pick the method by the question, then check the estimand

Where to go next

1. `install.packages("xplainfi")`
2. Documentation: https://mlr-org.github.io/xplainfi
3. `xplainfi` and practical guidance:

   xplainfi: Feature Importance and Statistical Inference for Machine Learning in R

   Burk et al. (2026), arXiv:2603.15306
4. Methodological background:

   A Guide to Feature Importance Methods for Scientific Inference

   Ewald et al. (2024), Explainable Artificial Intelligence (xAI 2024)

Thank you for your attention!

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Backup

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